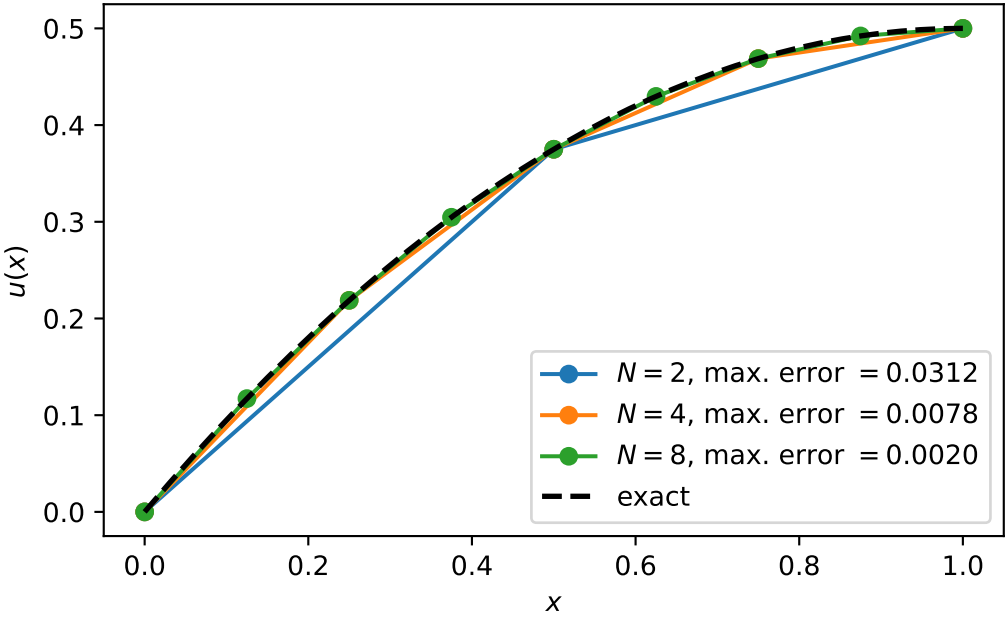
0.5
0.4
0.3
0.2
0.1
0.0
$u(x)$
0.0
0.2
0.4
0.6
0.8
1.0
$x$
$N = 2$, max. error $= 0.0312$
$N = 4$, max. error $= 0.0078$
$N = 8$, max. error $= 0.0020$
exact